A two-spectrum sextic criterion for Weinberg-model quartic stability

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Problem and claimed result

For three electroweak doublets define $K_{ij} = \Phi_i^\dagger \Phi_j$, $r_i = K_{ii}$, and

$$f(K) = \frac{1}{2}\sum_i a_i r_i^2 + \sum_{i<j}\left[b_{ij} r_i r_j + c_{ij}|K_{ij}|^2 + \mathrm{Re}(d_{ij}K_{ij}^2)\right]. \tag{1}$$

All a, b, c are real; all d are unrestricted complex numbers. Pair order is $12, 13, 23$. The normalized physical domain is

$$\mathcal{D} = \{K = K^\dagger \succeq 0 : \mathrm{tr}\, K = 1, \mathrm{rank}\, K \le 2\}. \tag{2}$$

Rank one is neutral; rank two is charge breaking. The central geometric object is the **rank-one projector onto a charged configuration's kernel**. It turns charged stationarity into a second instance of the neutral phase problem.

Claim. The minimum of (1) on (2) is obtained from two explicit spectra, each containing three singleton values, at most six pair values and at most six full-support values generated by a sextic. A scalar purity test selects physical charged candidates. An ordered positive infinitesimal completion makes the criterion valid for every coupling, including singular and flat-boundary cases, and recovers a minimizing field direction. No field-orientation optimization remains. This is a root/sign criterion, not a short list of polynomial inequalities or a claimed production-ready exact scanner. The sextic bound concerns phase elimination; residue annihilators need not have degree at most six.

The common spectrum: definition and explicit roots

For real α, e, g, complex d, unit-circle coordinates $q_1 = 1, q_2, q_3$, and $r_i \ge 0$, $\mathbf{1}^T r = 1$, define

$$B_{ii} = \alpha_i/2, \qquad B_{ij} = \tfrac{1}{2}[e_{ij} + \mathrm{Re}(d_{ij}q_j\bar{q}_i)], \qquad Q = \frac{r^T B r}{(g^T r)^2}, \quad g^T r \neq 0. \tag{3}$$

All signs of $g^T r$ are allowed. Each retained candidate carries its value $\nu = Q$, amplitudes r, phases q and projector $P_{ij} = \bar{w}_i w_j$, where $w_i^2 = r_i q_i$. Square-root sign copies have the same values and admissibility.

Singletons give $r = e_i$, $\nu = \alpha_i/(2g_i^2)$. For each pair $i < j$ and $\sigma = \pm 1$, put

$$q_j\bar{q}_i = \sigma\bar{d}_{ij}/|d_{ij}|, \quad A_i = \alpha_i/2, \quad A_j = \alpha_j/2, \quad h = (e_{ij} + \sigma|d_{ij}|)/2,$$
$$n = (A_j g_i - h g_j, A_i g_j - h g_i), \quad W = A_j g_i^2 - 2h g_i g_j + A_i g_j^2,$$
$$r_J = n/(n_i + n_j), \qquad \nu = (A_i A_j - h^2)/W. \tag{4}$$

Keep nonzero denominators, strictly positive supported amplitudes and $g^T r \neq 0$. Do not divide by ν or by $\det B_J$.

For full support set

$$s = (\mathrm{Im}(d_{12}q_2), \mathrm{Im}(d_{13}q_3), \mathrm{Im}(d_{23}q_3\bar{q}_2)), \qquad j = (s_{23}, -s_{13}, s_{12}).$$

When $\mathrm{Im}(d_{12}d_{23}\bar{d}_{13}) \neq 0$, phase stationarity implies nonzero conserved current $r_1 r_2 s_{12} = r_2 r_3 s_{23} = -r_1 r_3 s_{13}$, hence

$$r = j/(\mathbf{1}^T j). \tag{5}$$

Define

$$C = \begin{pmatrix} e_{13}d_{12} & -e_{12}d_{13} & \alpha_1 d_{23} - d_{13}\bar{d}_{12} \\ e_{23}d_{12} & d_{12}d_{23} - \alpha_2 d_{13} & e_{12}d_{23} \\ \alpha_3 d_{12} - d_{13}\bar{d}_{23} & -e_{23}d_{13} & e_{13}d_{23} \end{pmatrix}, \quad z = (q_2, q_3, q_3\bar{q}_2)^T.$$

The key identity is

$$2Bj = \mathrm{Im}(Cz). \tag{6}$$

Thus radial stationarity is reduced to $\mathrm{Im}(hz) = \mathrm{Im}(kz) = 0$, with $h = g_1 C_{2\bullet} - g_2 C_{1\bullet}$, $k = g_1 C_{3\bullet} - g_3 C_{1\bullet}$.

For a row (A, B, C), form

$$\begin{aligned} P_{A,B,C}(t,u) = {} & [\operatorname{Im} A(1-t^2) + 2\operatorname{Re} A\, t](1+u^2) \\ & + [\operatorname{Im} B(1-u^2) + 2\operatorname{Re} B\, u](1+t^2) \\ & + \operatorname{Im} C[(1-t^2)(1-u^2) + 4tu] \\ & + 2\operatorname{Re} C[u(1-t^2) - t(1-u^2)]. \end{aligned} \tag{7}$$

Use $F = P_h$, $G = P_k$ and

$$S(t) = \operatorname{Res}_u(F, G)/(1+t^2), \qquad \deg S \le 6. \tag{8}$$

The fixed factor comes from the nonreal common points $(i, i), (-i, -i)$. Recover every common real root of the specialized quadratics, using all four projective half-angle charts, and set $q_2 = (1 - t^2 + 2it)/(1 + t^2)$, similarly for q_3. Retain only positive (5) and nonzero $g^T r$. Coprime bidegree-(2,2) forms have eight intersections counting multiplicity; the fixed nonreal pair leaves at most six physical roots.

Primal and charged spectra

Let A be the real symmetric matrix with diagonal a_i and off-diagonal b_{ij}. The self-adjoint operator

$$(LX)_{ii} = (A \operatorname{diag} X)_i, \qquad (LX)_{ij} = c_{ij} X_{ij} + \bar{d}_{ij} \bar{X}_{ij}$$

satisfies $f(K) = \operatorname{tr}(KLK)/2$. The **primal spectrum** uses $(\alpha, e, d, g) = (a, b + c, d, \mathbf{1})$; every retained projector is physical and contributes its value ν.

For the second spectrum set $R = L^{-1}$, $G = A^{-1}$, $\Delta_{ij} = c_{ij}^2 - |d_{ij}|^2$, $g = G\mathbf{1}$, $\beta = \mathbf{1}^T G \mathbf{1}$. Then

$$(RX)_{\mathrm{diag}} = G \operatorname{diag} X, \quad (RX)_{ij} = (c_{ij} X_{ij} - \bar{d}_{ij} \bar{X}_{ij})/\Delta_{ij}.$$

The **dual spectrum** uses

$$\alpha'_i = 2G_{ii}, \quad e'_{ij} = 2(G_{ij} + c_{ij}/\Delta_{ij}), \quad d'_{ij} = -2d_{ij}/\Delta_{ij}, \quad g' = G\mathbf{1}. \tag{9}$$

For every retained P, define

$$\begin{aligned} a_P = \operatorname{tr}(PRI), \quad p_P = \operatorname{tr}(PRP), \quad D_P = \beta p_P - a_P^2, \\ K_P = (p_P RI - a_P RP)/D_P, \qquad E_P = p_P/(2D_P). \end{aligned} \tag{10}$$

Keep $D_P \neq 0$. Weighted stationarity gives $K_P P = 0$ and trace one. Consequently K_P is physical exactly when

$$\chi_P = \frac{p_P^2 \operatorname{tr}[(RI)^2] - 2p_P a_P \operatorname{tr}(RI\, RP) + a_P^2 \operatorname{tr}[(RP)^2]}{D_P^2} \le 1, \tag{11}$$

Equivalently compute $\chi_P = \operatorname{tr}(K_P^2)$ directly. A Hermitian trace-one matrix with a kernel has eigenvalues $(0, x, 1-x)$, proving the test. Include E_P for all passing candidates, not just the minimum of the auxiliary quotient p_P/a_P^2.

The coverage proof starts from charged stationarity $LK = \lambda I + \mu P$, independently obtained from the doublet equation $\Phi LK = \lambda \Phi$. Trace and kernel expectations give $\beta\lambda + a_P\mu = 1$, $a_P\lambda + p_P\mu = 0$, yielding (10) and $\lambda = 2f$. Its kernel equation is precisely projective stationarity of p_P/a_P^2. Sparse kernel supports are included by (4) and the singletons; the symmetry makes off-support derivatives vanish.

All exceptional inputs and the exact stability test

For generic coefficients, the phase forms are coprime and no charged stationary point has $a_P = p_P = D_P = 0$. The latter is a real possibility, not excluded by invertibility or a nonreal phase alone. Its generic exclusion is proved by restricting p_P to the pole locus $a_P = 0$. On support size s, that locus has dimension $2s - 3$, while its stationary-zero condition has $2s - 2$ independent value/first-jet constraints. The weighted inverse parameters are independent: their linear map from the fifteen structured inverse-operator coordinates has nonzero determinant 8192. These facts yield a proper algebraic exceptional set; an explicit rational coprime example proves the corresponding phase exception is proper.

To cover every original coupling, in coordinate order $(a, b, c, \operatorname{Re} d_{12}, \operatorname{Im} d_{12}, \dots, \operatorname{Im} d_{23})$, let M_j be its coefficient monomial and use independent ordered infinitesimals $0 < \epsilon_{15} \ll \cdots \ll \epsilon_1 \ll \epsilon_0 \ll 1$:

$$f_\epsilon = f + \epsilon_0(\operatorname{tr} K)^2 + \sum_{j=1}^{15} \epsilon_j M_j(K). \tag{12}$$

This is algebraic computation in an iterated real closed Puiseux field, not sampling at small real numbers. The deformed coefficients avoid every proper algebraic exception. If m_ϵ is the smallest retained physical value from the two spectra, then

$$|m_\epsilon - m - \epsilon_0| \le \tfrac{9}{2}\epsilon_1, \qquad m = \min_{\mathcal{D}} f. \tag{13}$$

The bound follows from $|M_j| \le 1/2$ for the three diagonal terms and $|M_j| \le 1/4$ for the twelve others. Therefore

$$\boxed{f \ge 0 \iff m_\epsilon > 0, \quad f > 0 \iff m_\epsilon > 2\epsilon_0, \quad m = \operatorname{st}(m_\epsilon)}. \tag{14}$$

The boundary has $0 < m_\epsilon < 2\epsilon_0$; outside the cone $m_\epsilon < 0$. All comparisons are exact ordered-field comparisons.

A minimizing matrix is bounded, so its successive residues, taken in the order $\epsilon_{15}, \dots, \epsilon_0$, preserve PSD, trace, rank and its minimum value. Each selected bounded algebraic root has a finite residue procedure: compute an annihilating polynomial, divide its common infinitesimal power, specialize, order its distinct real roots, and use midpoint comparisons to select the correct residue. Univariate Thom encodings implement roots/signs; no original field quantifier remains. This handles coalescing roots and removable poles. A Schur-complement Gram factor of the limiting matrix gives an exact negative or zero doublet direction.

Verification interface and limitations

The full proof is in `THEOREM.md`; exact counterexamples and alternative representations are in `DERIVATIONS.md`. Executed checks include arbitrary-symbolic (6) and (8), twelve field-gradient identities, exact singular/purity/infinity negative controls, an independently derived matching sextic, and a complete benchmark minimum of $-7/60$. Twelve floating sextic minima agree with separate field minimization records within 7.22×10^{-16}; that agreement is not a global certificate. No proof assistant, external reviewer or exact end-to-end software implementation has certified this work.

The strongest critical-review targets are the inverse-spectrum equivalence, full-support current elimination, pole-zero genericity argument, and the ordered-field-to-original-field limiting bridge. Tree-level quartic positivity does not settle arbitrary quadratic terms on a flat direction, the physical vacuum, or radiative stability.

Context and mathematical inputs

Jurčiukonis–Lavoura–Milagre, arXiv:2603.23590v3 (11 June 2026), gives the recent necessary-condition/ML baseline. General Gram/stationary methods are prior work: Maniatis–Nachtmann, arXiv:1408.6833, with its 2015 erratum. Generic resultant methods are also prior work: Ivanov–Köpke–Mühlleitner, arXiv:1802.07976. The claimed increment here is the explicit two-sextic completion, not generic decidability.

Real-root/sign and semialgebraic tools: Basu–Pollack–Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed. (2006). Ordered bounded-residue conventions: Lerario–Rizzi–Tiberio, *Quantitative approximate definable choices*, Math. Ann. 392 (2025), 1289–1319. The selected-root residue procedure is proved in the full text rather than imported from a simple-zero-only multivariate algorithm.